



ORION NanoFab User Guide

Generated from the maintained Markdown documentation.

Controlled source: online/Markdown documentation. Exported copies may become outdated.

Before You Start

Complete the pre-use checks before beginning a work session.

WARNING Stop conditions Do not proceed if there are unexplained errors, gun temperature is above about 84 K, less than 2 h LN $\blacksquare$ remains, or vacuum is abnormal. Contact a Superuser.

- 1 Cool down body/clothing temperature before entering.
- 1 Open System Manager, ZEN (after the server starts), and Tools → Vacuum Screen.
- 1 Check error messages and the logbook; check trimer status.
- 1 Check gun temperature (local SOP target < ~80 K).
- 1 Check LN $\blacksquare$ remaining time; do not proceed if <2 h.
- 1 Check vacuum status before loading.

Local SOP: Before Measurement.

Sample Loading & Unloading

Use Sample Management to exchange the holder through the air lock.

- 1 Select Exchange .
- 1 If the holder is in the chamber, select Transfer , insert the transfer rod steadily, fully couple it to the holder, then retract.
- 1 Keep the transfer rod fully retracted before Store ; verify the Rod Retracted interlock.
- 1 Select Vent , open the load lock and mount the sample. Photograph the holder position when useful for the Sample Map.
- 1 Close the load lock and return to Store . Wait for the load chamber to pump before Transfer.

NOTE Stage caution Use the chamber camera and established height limits to avoid collisions with the column or detector.

Logic review — item requiring confirmation

PLACEHOLDER — VERIFY STAGE CLEARANCE: The source SOP includes a stage/holder clearance condition during transfer, but the local documents use different wording and calibration values. Confirm the current safe transfer position/clearance and add it here as an explicit pre-transfer verification.

Starting a Session

- 1 Confirm the appropriate aperture before turning on the gun.
- 1 Power the ET detector; power the flood gun only when needed.
- 1 Select helium and allow gas pressure to stabilize.
- 1 Set landing energy, aperture and Spot Control for the intended work.
- 1 Enable Lens 2 and focus according to the local SOP/training.

WARNING Operator boundary Do not use Extract Voltage as an ordinary user control unless explicitly trained/authorized.

Trimer Check & Alignment

- 1 Move to the standard Au sample / saved reference site.
- 1 Rough focus at the documented working distance and use a short dwell time.
- 1 Enter SFIM mode and grab an image.
- 1 Check whether the designated trimer atom is centered; use Gun Tilt if adjustment is required.
- 1 Expose the trimer as briefly as possible, then return to normal imaging/Gun Shift mode.

Imaging

Tune on a standard sample where possible to minimize exposure of the specimen.

- 1 Navigate to the region of interest and use Continuous scanning for setup.
- 1 Set the desired beam current/spot, dwell time and scan size.
- 1 At a flat region, iteratively correct astigmatism, Gun Shift/wobbler and focus.
- 1 For capture, the local SOP suggests line averaging 8/16 and 2 μ s dwell; increase dwell when higher quality is required.
- 1 Stop scanning, then use Grab and save the image to the approved data location.

Patterning

- 1 Open the Patterning tab and focus before patterning.
- 1 Set dose in ions/cm² and verify expected beam current.
- 1 Use the local SOP values appropriate to He or Ne and the intended process.
- 1 For complex patterns, use NPVE; verify the server/communication connection before starting.
- 1 Draw/import patterns, review parameters, select the intended patterns and start from the NPVE GUI.

NOTE Source note The local SOP records example process values; treat them as local working notes, not universal defaults.

Logic review — item requiring confirmation

PLACEHOLDER — VERIFY PATTERNING PARAMETERS: The local SOP itself contains “Other patterning parameters?” after the listed dose/current/dwell/pixel-spacing guidance. Define the approved parameter checklist for He, Ne, ZEN and NPVE before treating this page as a complete patterning procedure.

Working with Neon

- 1 Set up the microscope and save stage positions while still working with helium.
- 1 Switch to a neon-compatible aperture before changing gas.
- 1 Select Neon in Gas Control and wait for helium evacuation and neon regulation.
- 1 Proceed with the normal GFIS workflow and re-check alignment.

WARNING Aperture warning The ZEISS manual states gold apertures are easily damaged by neon; use the designated neon/moly aperture positions for the installed aperture set.

Ending a Session

- 1 Return tilt and scan rotation to 0°.
- 1 Select Exchange to move to the transfer condition.
- 1 Unload the sample using Transfer → Vent → Store, or load the next sample.
- 1 Make sure gas is Off / GFIS is in Standby.
- 1 Power down the detector as required; return the standard holder/sample according to the local SOP.
- 1 Complete the logbook, clean the work area and transfer data using the approved method.

Warning Signals / When to Stop

WARNING Stop and contact a Superuser Unexplained system errors; gun temperature above the permitted range; insufficient LN■; abnormal vacuum; O■/fire/facility alarms; unexpected mechanical or transfer behavior.

Normal users should recognize abnormal states but should not attempt chamber venting, high-voltage work, pump recovery, source maintenance or power-outage recovery.